

#HW 3

● Graded

Student

Victoria Worotniak

Total Points

100 / 100 pts

Question 1

Q1

10 / 10 pts

✓ - 0 pts Correct

- 3 pts Your calculation and graph is not clear

Question 2

Q2

10 / 10 pts

✓ - 0 pts Correct

- 2 pts There is a mistake in calculation

- 4 pts Incomplete

Question 3

Q3

10 / 10 pts

✓ - 0 pts Correct

- 4 pts not clear

- 5 pts incomplete

Question 4

Q4

10 / 10 pts

✓ - 0 pts Correct

- 2 pts error in calculation

- 5 pts incomplete

Question 5

Q5

14 / 14 pts

✓ - 0 pts Correct

- 2 pts Final pressure is wrong.

- 4 pts unclear

- 4 pts incomplete

- 2 pts unit conversion not done.

Question 6

Q6

46 / 46 pts

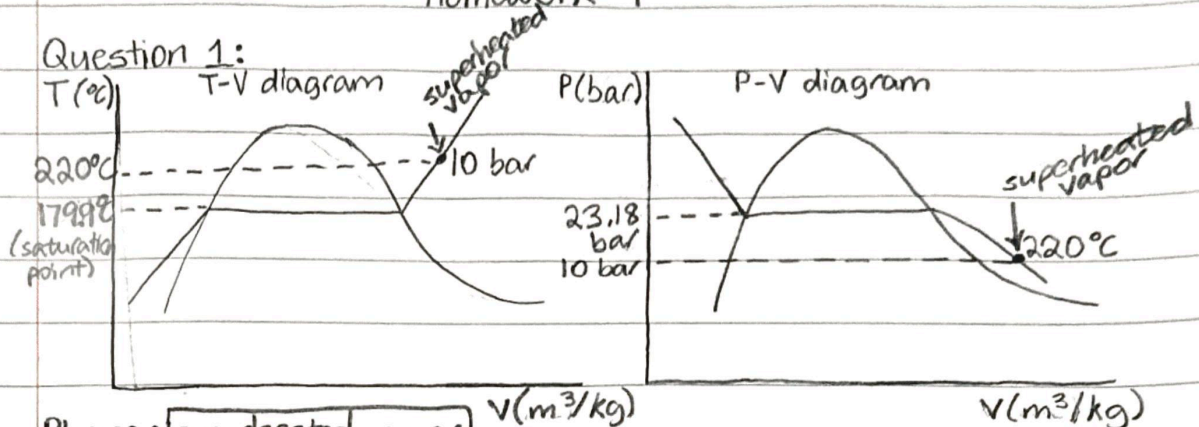
✓ - 0 pts Correct

- 0 pts [Click here to replace this description.](#)

Questions assigned to the following page: [1](#), [2](#), and [3](#)

Homework 4

Question 1:

Phase: superheated vapor

$$200^\circ\text{C} + 10 \text{ bar}: 0.2060 \text{ m}^3/\text{kg} \quad v = 0.2060 + \frac{220-200}{240-200}(0.2275-0.2060)$$

$$240^\circ\text{C} + 10 \text{ bar}: 0.2275 \text{ m}^3/\text{kg}$$

$$v = \boxed{0.217 \text{ m}^3/\text{kg}}$$

$$200^\circ\text{C} + 10 \text{ bar}: 2621.9 \text{ kJ/kg} \quad u = 2621.9 + \frac{220-200}{240-200}(2692.9-2621.9)$$

$$240^\circ\text{C} + 10 \text{ bar}: 2692.9 \text{ kJ/kg}$$

$$u = \boxed{2657 \text{ kJ/kg}}$$

Question 2:

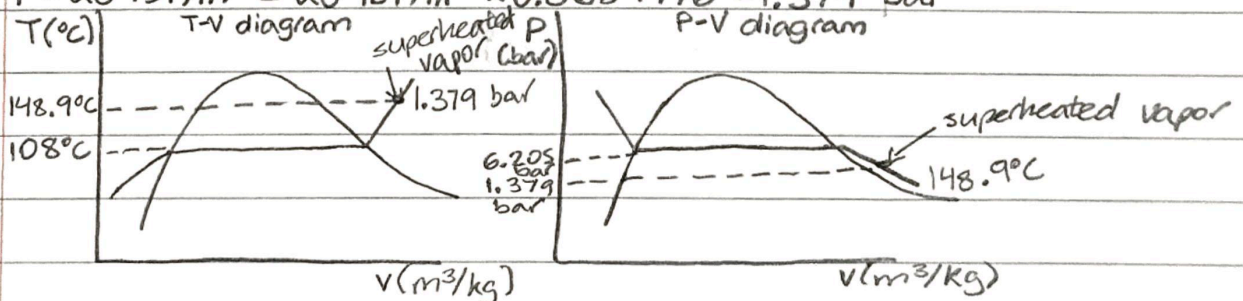
$$u = (1-x)u_f + xu_g \quad u = u_f - xu_f + xu_g \quad u - u_f = xu_g - xu_f$$

$$x = \frac{u - u_f}{u_g - u_f} = \frac{300 \text{ kJ/kg} - 251.3 \text{ kJ/kg}}{478.7 \text{ kJ/kg} - 251.3 \text{ kJ/kg}} = 0.214 \text{ or } \boxed{21.4\%}$$

Question 3: (Problem 3.21.8)

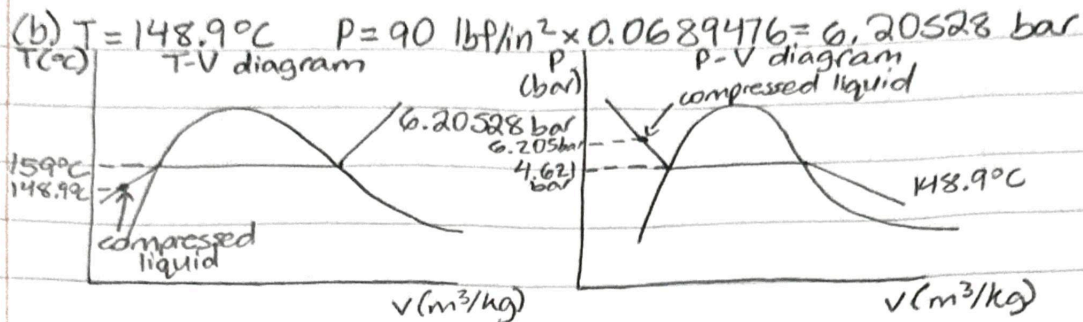
$$(a) T = 300^\circ\text{F} = (300^\circ\text{F} - 32) \cdot \frac{5}{9} = 148.9^\circ\text{C}$$

$$P = 20 \text{ lbf/in}^2 = 20 \text{ lbf/in}^2 \times 0.0689476 = 1.379 \text{ bar}$$

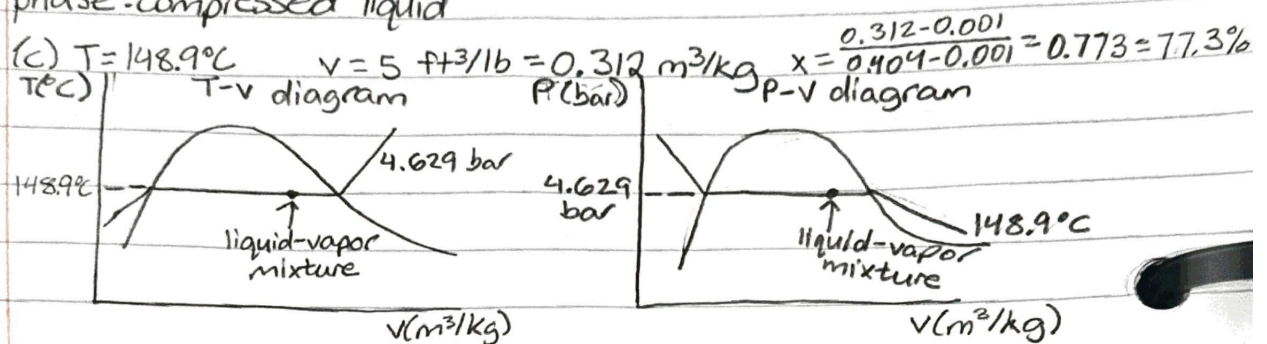


phase: superheated vapor

Questions assigned to the following page: [3](#), [4](#), and [5](#)



phase: compressed liquid



phase: liquid-vapor mixture

Question 4: (3.21.11)

$T = 40^\circ\text{F}$ $v_f = 0.01250 \text{ ft}^3/\text{lb}$ $m = 2 \text{ lb}$

$P_{\text{sat}} = 49.771 \text{ psia}$ $v_g = 0.9523 \text{ ft}^3/\text{lb}$ $x = 0.20$

$v_{\text{mix}} = 0.01250 + 0.20(0.9523 - 0.01250) = 0.20 \text{ ft}^3/\text{lb}$

$V = m v_{\text{mix}} = (2 \text{ lb})(0.20 \text{ ft}^3/\text{lb}) = 0.401 \text{ ft}^3$

Since it is saturated, at 40°F $P = 49.77 \text{ psia}$

Question 5: (3.21.15)

Closed rigid tank $\rightarrow v$ is constant

Initial: $T_1 = 200^\circ\text{F}$ $x_1 = 0.30$ (two-phase)

Final: $T_2 = 240^\circ\text{F}$

Question assigned to the following page: [5](#)

At 200°F:

$$P = 11.538 \text{ psia}$$

$$V_f = 0.01839 \text{ ft}^3/\text{lb}$$

$$V_g = 33.609 \text{ ft}^3/\text{lb}$$

$$V = (1-x)V_f + xV_g = (1-0.30)0.01839 + 0.3(33.609) = 10.096 \text{ ft}^3/\text{lb}$$

At 240°F:

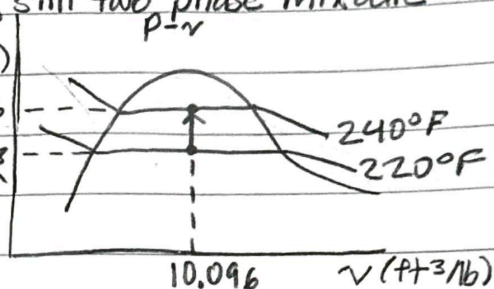
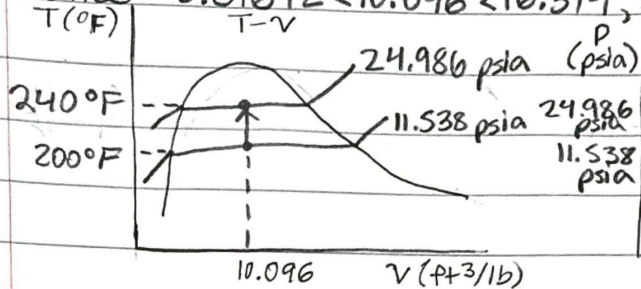
$$P = 24.986 \text{ psia}$$

$$V_f = 0.01692 \text{ ft}^3/\text{lb}$$

$$V_g = 16.314 \text{ ft}^3/\text{lb}$$

$$x_2 = \frac{10.09 - 0.017}{16.314 - 0.017} = 0.6188$$
$$= 61.9\%$$

Since $0.01692 < 10.096 < 16.314$, still two phase mixture



phase: liquid-vapor mixture for both

Question assigned to the following page: [6](#)



CHE 201: Introduction to Thermodynamics

Expires Dec 29th, 2025



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HW 3 Question 6

46 / 46 points

Due: 09/22/2025, 11:59 PM CDT



3.1 Getting started

7 / 7 points



3.2 p-v-T relation

9 / 9 points



3.3 Studying phase change

9 / 9 points



3.4 Retrieving thermodynamic properties

5 / 5 points



3.5 Evaluating pressure, specific volume, and temperature

16 / 16 points